

Consumer impatience in the e-commerce home delivery context: A consumer-centric supply chain perspective for time-based competition

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Abstract

Purpose – This paper aims to introduce the concept of consumer impatience, empirically explore how it relates to time-based logistics performance (delivery speed and delivery timeliness), and discuss its impact on consumer satisfaction. This research argues that gaining insights related to delivery performance from a consumer’s perspective can help the development of more effective time-based logistics strategies for e-commerce home deliveries.

Design/methodology/approach – Hypotheses in this study are developed using attribution theory and tested with empirical data collected through an online behavioral consumer experiment. Middle-range theorising is used to develop an understanding of the mechanisms that impact the relationship between time-based logistics performance and consumer satisfaction.

Findings – Findings indicate that consumer impatience with delivery speed and delivery timeliness play an essential role in the relationship between time-based delivery performance and consumer satisfaction. Issues with delivery timeliness are shown to have a more negative impact on consumer satisfaction than issues with delivery speed, while delivery communication is demonstrated to have a positive relationship with consumer satisfaction.

Originality/value – This empirical study adds to existing time-based competition literature by taking a consumer-centric perspective and bringing a largely overlooked but critical concept – consumer impatience – into the logistics and supply chain management setting. Middle-range theorising allows for a conceptualized understanding of consumers’ delivery experiences that can help companies develop proactive actions in their time-based competition initiatives.

Keywords Consumer impatience, time-based competition, delivery speed, delivery timeliness, delivery communication

Paper Type Research Paper

Introduction

Managing supply chains has been described as “more challenging than ever,” partially due to rises in e-commerce and the escalating needs and expectations of consumers – the ultimate customers of a supply chain (Inbound Logistics, 2023). Consumers want fast deliveries, real-time information, and consistently high service at affordable prices (Solis, 2017; Drenik, 2024). Many companies have responded to increases in consumer demands by adopting aggressive logistics strategies (Esper *et al.*, 2020; Ghosh, 2023). In fact, fulfillment has been identified as the new

retailing battleground, highlighting how important it is for firms to understand evolving consumer expectations and needs (Gielens, 2023).

Consumer spending habits depend on their delivery experiences, and the consistently evolving nature of consumer preferences makes it critical that companies understand how their delivery services are perceived and valued (Drenik, 2024). Changing preferences mean that delivery experiences that were previously acceptable may no longer meet consumer expectations. Therefore, firms must develop a more nuanced understanding of the relationship between consumer time-based delivery experiences and satisfaction with delivery performance (Masorgo *et al.*, 2023; Olsson *et al.*, 2022). The current research contributes additional understanding to this relationship by exploring the role of *consumer impatience* in the context of e-commerce home deliveries. While satisfaction is the difference between expectations and actual experience (Oliver, 1980, 1993), impatience is the negative emotion during the waiting period for something positive (Hardisty and Weber, 2020). Our research focuses on consumer satisfaction with delivery performance relative to their perceived standard (Davis-Sramek *et al.*, 2009; Murfield *et al.*, 2017; Ta *et al.*, 2023). We believe it is imperative to develop a better understanding of consumer impatience since it can occur throughout the delivery process and can, therefore, negatively impact consumer satisfaction. A study reported by MH&L indicated that 46% of consumers feel impatient (MH&L Staff, 2021). Daugherty *et al.* (2019) also noted that we are now in the new age of impatience. However, there has been no empirical exploration of consumer impatience in the logistics performance context. Thus, we are interested in exploring the role of impatience in the relationship between delivery performance and consumer satisfaction. This research aims to introduce the concept of consumer impatience in a time-based

competition context and provide insight into its primary sources and subsequent consumer satisfaction.

We argue that studying the topic of consumer impatience will allow logistics and supply chain management (LSCM) researchers and practitioners to better integrate the increasingly critical consumer perspective into time-based competition strategies. By exploring consumer impatience, our research aims to enhance understanding of specific aspects of delivery performance that can be leveraged to provide the most significant benefits to companies and their customers. Although LSCM scholars have used modeling techniques to explore interactions between impatience, consumer willingness to wait for goods or services, and consumer behavior (e.g., Benjaafar *et al.*, 2010; Kumar and Sharma, 2012), it is still unclear how the specific attributes of time-based delivery performance can affect the development of consumer impatience and subsequent satisfaction outcomes. While consumer impatience is likely to have a strong relationship with satisfaction, the nature of this relationship deserves investigation to provide valuable insights when developing appropriate e-commerce home delivery strategies. Existing literature suggests that communication impacts customer satisfaction (e.g., Duan *et al.*, 2022; Itani *et al.*, 2020; Jin *et al.*, 2023), but the mechanisms related to this effect have yet to be sufficiently elucidated. To further our understanding of this phenomenon, we argue that the effect of firm communication on consumer satisfaction could be due in part to its potential to mitigate the relationship between consumer impatience and overall satisfaction with a home delivery experience. Accordingly, the following research questions are proposed:

RQ1: *How does consumer impatience impact the relationship between time-based delivery performance and consumer satisfaction?*

RQ2: *How can firms use communication to alleviate the impact of consumer impatience on consumer satisfaction?*

The hypothesized relationships in this study were developed based on a combination of middle-range theorising and attribution theory. Middle-range theorising is a context-specific conceptualisation that provides theoretically grounded insights that are applicable to an empirical context (Ketchen *et al.*, 2022). Middle-range theorising complements general theorising since it can generate insights that are unique to the LSCM context (Malik and Ali, 2024; Stank *et al.*, 2017). Attribution theory (Heider, 1958; Weiner, 1986) is concerned with understanding how consumers perceive an undesirable or unexpected event's stability, controllability, and locus dimensions to determine blame for an issue (Folkes, 1984; Oflaç *et al.*, 2012). A scenario-based experiment methodology was chosen for this research since it is ideal for investigating events within a specific context (Jones *et al.*, 2023). In our experiments, consumers were exposed to scenarios containing manipulations related to delivery speed and delivery timeliness. Consumer impatience associated with each time-based logistics performance manipulation was measured. Next, consumers were exposed to communication manipulations, with the goal of understanding the potential moderating impact of communication on the relationship between impatience and consumer satisfaction with delivery performance.

A contribution of this study is the introduction of consumer impatience in the LSCM setting through empirical data collection and analysis. The role of consumer impatience is explored by middle-range theorising (Craighead *et al.*, 2024; Stank *et al.*, 2017), which allows for the examination of the how's, why's, and when's in the well-established relationship between time-based delivery performance and consumer satisfaction. Our research can help the effort to integrate consumers' perspectives into LSCM research and practices (Peinkofer *et al.*, 2022; Masorgo *et al.*, 2023) by examining not only satisfaction related to time-based deliveries but also by bringing attention to the important psychological factors that play a critical role in this

relationship. More specifically, we introduce consumer impatience as a construct critical to better understanding logistics performance and time-based competition. Studying consumer impatience within a supply chain context provides the opportunity for an improved understanding of how logistics services can be leveraged to meet consumer expectations and maximize consumer satisfaction.

Literature background

Time-Based Competition and Consumer-Centric SCM

Time has long been considered a powerful source of competitive advantage (Stalk, 1988), where many supply chain performance metrics (e.g., order lead time, on-time delivery, cash-to-cash cycle, etc.) are time-based (Stalk and Hout, 1990; Thomas, 2008; Thomas *et al.*, 2014).

Recognizing the strategic role of time and focusing on shrinking the time requirements related to critical business activities can yield a competitive advantage and lead to exceptional firm performance (Hult *et al.*, 2002; Srivastava *et al.*, 1999). More specifically, time-based competition strategies that emphasize customer responsiveness and supply chain time compression can yield substantial benefits, including increased brand loyalty, extended market share, increased price premiums, and improved cost efficiency (Bozarth and Chapman, 1996; Stalk and Hout, 1990; Thomas, 2008).

Consumers' perspectives are highly relevant when developing time-based strategies because they are the ultimate judge of companies' performance. Thus, LSCM scholars have argued that it is imperative to integrate a consumer-centric perspective into time-based competition strategies to maximize their effectiveness (e.g., Esper and Peinkofer, 2017; Rich and Hines, 1997). Previous research within the e-commerce context has reasoned that consumers are especially sensitive to issues related to timeliness and delivery speed (Nguyen *et al.*, 2019).

While it is true that servicing and satisfying the demands of end-consumers should be the ultimate goal of the entire supply chain, additional research is needed to explore consumer preferences and perceptions related to the logistics services that they receive (Esper *et al.*, 2020; Frankel *et al.*, 2008).

Efforts to better understand the LSCM activities required to provide a satisfactory consumer experience have been increasing in recent years (e.g., Kumar *et al.*, 2023; Masorgo *et al.*, 2023; Mollenkopf *et al.*, 2022; Song *et al.*, 2024; Tokar *et al.*, 2020). Traditional LSCM performance measures tend to be internal, focusing on financial and operational outcomes that often lack consumer input (Fugate *et al.*, 2010; Sanders *et al.*, 2013). In their structured literature review, Esper and Peinkofer (2017) noted that consumers' reactions, perceptions, and assessments of LSCM performance were rarely considered in existing research. This omission can significantly hinder companies' efforts in developing and implementing successful LSCM strategies because they lack sufficient understanding of what their customers value (Esper *et al.*, 2020). Our study contributes to the existing body of work by deepening our understanding of the relationship between time-based delivery performance and consumer satisfaction. Table A in the supplementary material highlights recent work in this domain and suggests how our work meaningfully extends these efforts. Our research contributes to existing studies by considering both timeliness and speed simultaneously in our empirical investigation of time-based delivery performance, allowing us to gain a better understanding of the relationships between these two factors. Additionally, our study contributes to existing literature that tends to focus on consumer outcomes such as satisfaction or loyalty. Instead, we take a more nuanced approach using middle-range theorising to examine consumer psychological factors, such as impatience, that impact the relationship between time-based delivery performance and consumer satisfaction.

Through middle-range theorising, we intend to develop a better understanding of *why* time-based delivery performance impacts consumer satisfaction.

Consumer Impatience

Previous literature has shown that time-based logistics performance is related to consumer satisfaction (Esper and Peinkofer, 2017), and we argue that impatience is a critical yet overlooked element in understanding the intricacies of this relationship. Given their common roots in consumer psychology, impatience and satisfaction must first be clearly distinguished.

Impatience is the negative emotion associated with the anticipation of a positive future event (Hardisty and Weber, 2020). Impatience occurs before the receipt of a delivery. In contrast, *satisfaction* is an overall judgment of performance based on the difference between expectations and realized performance (Oliver, 1980, 1993). It is applicable after receiving a delivery – encompassing a retrospective and holistic view of a consumer’s experience.

Consumer impatience related to logistics performance is tied to the ability of supply chains to develop and execute effective time-based competition strategies. The increasing frequency and severity of consumer impatience are likely related to several LSCM performance-related factors. First, delays in service recovery should increase consumer impatience because it impedes consumers’ ability to realize the positivity associated with order delivery (Russo *et al.*, 2022). For example, impatience due to a stockout during a traditional in-store shopping trip will be minimized if the store offers alternative product options. These alternatives are less readily available when delivery issues arise in online retailing situations, likely increasing the negative emotions associated with anticipation and ultimately leading to higher impatience levels. Second, the nature of online retailing increases consumers’ exposure to delivery issues compared to traditional in-store retailing settings. When waiting for the arrival of an online delivery, several

problems could occur – such as late or slow deliveries – that result in heightened impatience. These events are less likely to occur during traditional in-store transactions. In fact, Tea and Robinson (2021) noted that the combination of recent supply chain disruptions and a surge in online shopping has led to overstretched companies and impatient customers. Third, the impersonal and self-service nature of online shopping can have significant negative impacts on consumer experiences (Dadzie *et al.*, 2005). As consumers continue to shift toward e-commerce, they will continue to become more disconnected from retailers and, therefore, less understanding of delivery-related issues. The increasingly transactional relationship between consumers and online retailers will likely increase consumer impatience.

Studies outside of the field of LSCM have previously tied impatience to timing and information availability (e.g., Chen *et al.*, 2005; DeVoe and House, 2012 Hardisty and Weber, 2020; Shaddy and Lee, 2020), but impatience has not yet been empirically studied within a logistics setting. Although previous research provides guidance related to general consumer preferences, it is important for LSCM practitioners to understand the different logistics-related drivers of impatience. Gaining such knowledge should allow for the development of more effective time-based delivery strategies to maximize consumer satisfaction.

While consumer impatience within the context of e-commerce or omni-channel retailing has received little empirical attention in LSCM literature, existing research has explored related topics. For example, LSCM researchers have explored customer service and logistics performance related to e-commerce and omni-channel retailing (Adams *et al.*, 2014; Gligor *et al.*, 2020; Jones *et al.*, 2023; Mangiaracina *et al.*, 2019; Risberg *et al.*, 2023). LSCM research has also studied the relative value consumers place on delivery speed, return and exchange policies, product and service quality, supply chain transparency, and online versus in-store retailing

flexibility (Gawor and Hoberg, 2018; Ma, 2017; Marino *et al.*, 2018; Song *et al.*, 2020). Our study extends this stream of research by leveraging and extending these findings to further our understanding of the relationships that exist between time-based logistics performance, consumer impatience, and consumer satisfaction.

Theory and hypothesis development

Attribution theory asserts that making attributions is a critical component of processing information (Heider, 1958; Weiner, 1986). When adverse or unexpected events occur, consumers tend to conduct attributional searches related to causality to determine who or what was to blame for the undesirable outcome (Folkes, 1984; Hartmann *et al.*, 2022). In the current research, the negative and unexpected events are associated with time-based delivery performance. While Hardisty and Weber (2020) note that the positive emotions associated with waiting for a positive event (i.e., savoring) typically accompany the negative emotion of impatience, we believe that in cases of a delivery failure related to delivery speed or delivery timeliness, consumer impatience will substantially outweigh the positive feeling related to savoring. Further, we contend that the attributions consumers make when experiencing impatience are critical in determining consumer satisfaction.

In attribution theory, causality attributions have three primary dimensions (Heider, 1958; Folkes, 1984): (1) stability, the perception of causes as temporary or permanent; (2) controllability, the belief as to whether an actor could impact or prevent the occurrence of an issue; and (3) locus, which determines if a cause is internal or external to an actor. Stable, controllable, and internal judgments are likely to lead to a causal attribution associated with the character of a company. In contrast, isolated, uncontrollable, and external judgments are more likely to lead to a more contextual causal attribution for the outcome. We expect consumers to

view delivery performance issues such as late or slow deliveries as controllable and internal. Consumers' impatience with these issues should result in decreased levels of consumer satisfaction related to a delivery experience.

While delivery speed is defined as how quickly a delivery is made after an order has been placed, delivery timeliness refers to the ability of a firm to deliver a product or service on time as promised (Bienstock *et al.*, 1997; Nguyen *et al.*, 2019). Delivery speed is related to delivery time and order lead time (order cycle time). Griffis *et al.* (2012) explored the relationship between delivery speed and consumer satisfaction, finding order cycle time to be positively related to purchase satisfaction. In a similar study, Rao *et al.* (2011) detailed the positive relationship between physical distribution service quality (which includes availability as a critical element) and customer purchase satisfaction. Although previous research has explored the relationship between delivery speed and satisfaction, we believe impatience is a critical (and largely overlooked) mediator for this relationship.

We assert that impatience will substantially increase if a consumer experiences long lead times for their e-commerce home deliveries. Since consumer impatience is the negative emotion associated with waiting for a positive event (Hardisty and Weber, 2020), slow deliveries should increase the levels of impatience that are experienced. In turn, the negative emotions that accompany impatience are likely to trigger an attribution process that results in consumers viewing slow deliveries as stable, controllable, and internal. Due to these attributions, consumers are likely to blame retailers for the long lead times, resulting in lower levels of satisfaction with delivery performance. Therefore, we hypothesize:

H₁: *Consumer impatience with delivery speed mediates the relationship between delivery speed performance and consumer satisfaction.*

Similarly, attribution theory can be applied to hypothesize the relationship between delivery timeliness and satisfaction. Delivery timeliness has consistently been shown to have a positive relationship with consumer satisfaction (Griffis *et al.*, 2012; Murfield *et al.*, 2017; Rao *et al.*, 2011). As delivery timeliness decreases, we expect higher levels of impatience related to the unexpected delay in the delivery process. In turn, the negative emotions accompanying this unexpected event will trigger an attribution process which is likely to again result in consumers attributing the issue to the retailer, which will negatively impact consumer satisfaction with delivery performance. As such, we hypothesize:

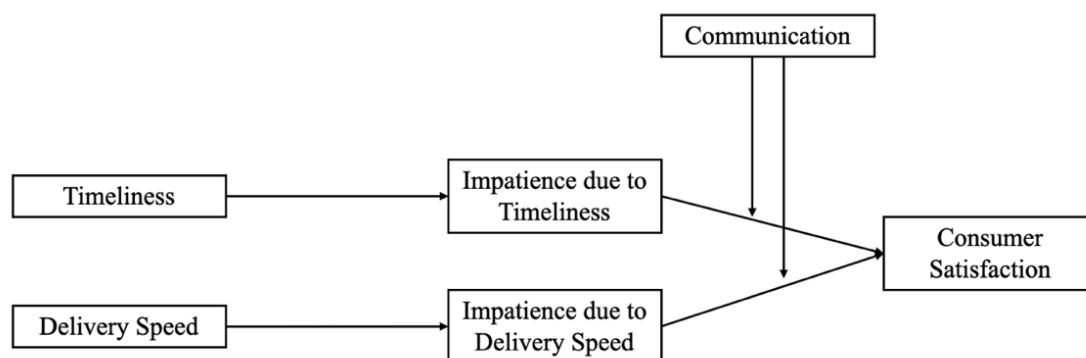
H₂: *Consumer impatience with delivery timeliness mediates the relationship between delivery timeliness performance and consumer satisfaction.*

Communication has a strong relationship with consumer satisfaction (e.g. Cho *et al.*, 2008; Daugherty *et al.*, 1998; Itani *et al.*, 2020; Mollenkopf *et al.*, 2022). We argue that the nature of the relationship between consumer impatience with logistics performance and consumer satisfaction should change as a function of firm delivery communication. Through communication, a firm can influence the attributions that the consumer makes regarding time-based delivery performance. For example, sharing specific information about unexpected issues that lead to a delay in processing a shipment can encourage consumers to view the delay as temporary and outside of the firm's control, increasing contextual attributions that are likely to bolster satisfaction with the firm's performance. Similarly, providing detailed tracking information when the package is shipped can help consumers view subsequent delays as beyond the control of the focal firm, which should help reduce negative character attributions (Heider, 1958; Folkes, 1984). In instances where consumers make contextual attributions for undesirable logistics performance, we expect the relationship between experienced impatience and subsequent satisfaction to be relatively modest because they are likely to attribute the source of

their frustration to actors outside the focal firm, leaving their satisfaction with the firm's delivery performance relatively intact. In contrast, to the extent that consumers make character attributions, they are likely to attribute poor performance to the focal firm. Accordingly, this attribution is likely to strengthen the relationship between impatience and satisfaction. Thus, we argue that delivery communication will moderate the relationship between consumer impatience with logistics performance (delivery speed and delivery timeliness) and consumer satisfaction.

H₃: *Delivery communication will moderate the negative relationship between consumer impatience with delivery speed and consumer satisfaction such that the relationship will be weaker when communication is high compared to when it is low.*

H₄: *Delivery communication will moderate the negative relationship between consumer impatience with delivery timeliness and consumer satisfaction such that the relationship will be weaker when communication is high compared to when it is low.*



Source(s): Figure created by the authors

Figure 1. Conceptual Model

Overview of research methodology

This research involves two experimental studies, each designed to test specific elements of the hypotheses. Pretests were conducted for both studies to ensure the efficacy of the experimental stimuli (Knemeyer and Naylor, 2011; Tokar, 2020). The studies were both designed to simulate an online purchase and delivery experience, where respondents were asked to reflect on their

experiences through a survey measuring impatience and satisfaction with their delivery experience.

Actions were taken to increase the internal validity of the experimental design for both studies. First, a between-subjects experimental design was selected to maximize internal validity via random assignment, greater control over conditions, and the manipulation of independent control variables (Confente *et al.*, 2021; Davis-Sramek *et al.*, 2018; Kirk, 2012; Knemeyer and Naylor, 2011). All scenarios utilized the same firm and product category to reduce the presence of product- or company-specific variation in expectations (Rao *et al.*, 2014). Accordingly, all participants were presented with the same baseline scenarios. To reduce social desirability concerns, respondents were told that there were no right or wrong answers and were asked to respond based on how they *would* react instead of how they *should* react. (Fischer 1993, Davis-Sramek *et al.*, 2018). To ensure the external validity of this experiment, each condition was designed to place participants into realistic and believable delivery situations. Simulated email messages and photos were embedded in each vignette to increase the level of immersion for each participant (Aguinis and Bradley, 2014; Wiedmer *et al.*, 2020). Before distribution, subject matter experts pretested each vignette for realism and clarity.

All participants were recruited using Amazon's Mechanical Turk (MTurk) service and received monetary compensation for their responses. Participants were required to be 18 or older and located in the United States. Any participant unable to commit to providing thoughtful responses during the process was excluded. Additionally, all participants were required to correctly answer two questions about vignette content, ensuring they were familiar with the situational characteristics. Directed response and reverse-coded questions were included throughout the surveys to check for thoughtful responses (Meade and Craig, 2012). Finally,

responses were evaluated for the total amount of time taken to complete the experiment (Bowling *et al.*, 2023).

Study 1

Experimental Design

A 2 (delivery speed: high vs. low) $\times$ 2 (delivery timeliness: high vs. low) $\times$ 2 (delivery communication: high vs. low) scenario-based experiment was utilized to test the proposed hypotheses. Participants were randomly assigned to one of the eight scenarios. Participants placed in a scenario with low (high) delivery speed were promised seven (one) day(s) between order placement and delivery. Participants in the low delivery timeliness condition experienced a two day delivery delay, while those in the high delivery timeliness condition received their package as promised. All participants viewed simulated emails that contained order confirmations, estimated delivery timing, and final delivery confirmation emails. Participants in the low delivery communication scenarios did not receive updates or delay notifications from the online retailer throughout their purchase transaction, while those in the high delivery communication scenarios were provided with regular updates. The scenarios and manipulations in this study are detailed in the supplementary material.

Consistent with the temporal nature of impatience, the experiment was designed to simulate the passage of time while participants waited for their orders to arrive. To enhance realism, we drew on research establishing that perceptions of the passage of time and the duration of an event are separate psychological processes (Jording *et al.* 2022). Specifically, to represent each day of waiting for a delivery to arrive, participants were required to navigate through a webpage depicting an image of a drive-through coffee provider and a textual description of their coffee acquisition routine for that simulated day. By varying the number of

such pages in accordance with the number of days participants' orders took to arrive, we were able to vary the objective time duration of the simulated wait time between conditions because participants had to navigate differing numbers of pages. In addition, because visual cues and repetitive tasks have both been shown to affect passage of time judgements (Ma *et al.*, 2024; Thönes *et al.*, 2018), including these design elements on each page allowed us to further enhance perceptions of elapsed time.

To ensure manipulation effectiveness, we included objective delivery performance manipulation checks for each scenario (Rungtusanatham *et al.*, 2011). Specifically, each participant was asked to rate the level of service that they received related to delivery communication, delivery speed, and delivery timeliness. Results of *t*-tests supported validity of the manipulations: delivery communication ($M_{\text{Low}} = 2.16$, $M_{\text{High}} = 5.25$; $t = -23.92$ ($df = 373$), $p < 0.001$); delivery speed ($M_{\text{Low}} = 2.75$, $M_{\text{High}} = 4.89$; $t = -16.97$ ($df = 323$), $p < 0.001$); and delivery timeliness ($M_{\text{Low}} = 2.16$, $M_{\text{High}} = 5.25$; $t = 29.88$ ($df = 371$), $p < 0.001$). These results confirm that participants could recognize the manipulated logistics performance differences across scenarios.

Sample

A total of 400 responses were collected through MTurk, with 50 responses collected per scenario. After the data were reviewed based on the attention checks, response variation, and the amount of time taken to complete the experiment, 24 participants were excluded from further analysis. The sample was 54.5% male; 81.3% identified their ethnicity as white. Much of the sample (80.3%) reported their age to be between 30 and 59 years. The annual income for 86.7% of participants was reported to be under \$100,000. The vast majority (93.8%) of participants

indicated that they make online purchases at least once per month, and all participants reported making at least one online purchase within the last year.

Measures

The primary outcome variable was consumer satisfaction with their scripted online purchase experience based on the logistics performance of the online vendor. All consumer satisfaction and impatience measures used a 7-point Likert scale, ranging from strongly disagree (1) to strongly agree (7). The specific scale items used are presented in Table B in the supplementary material. Consumer satisfaction was measured using four items adopted from Murfield *et al.* (2017). Consumer impatience was evaluated using measures created by adapting a general patience scale (Dudley, 2003) to incorporate customer expectations related to delivery speed and delivery timeliness. Impatience due to delivery speed and delivery timeliness were each measured using a four-item scale adapted from previous research (Bouzaabia *et al.*, 2013; Forslund, 2007; Mentzer *et al.*, 2001). Control variables included age, income, gender, and online purchase frequency. Descriptive statistics and correlations are summarized in Table C in the supplementary material.

Since the scales used to measure consumer satisfaction and impatience with logistics performance were adapted from previous scales, a CFA was used to test the proposed model. The fit of the specified CFA model was evaluated using multiple indices ($\chi^2 = 191.47$, $df = 51$, χ^2 p-value = 0.000, RMSEA = 0.086, SRMR = 0.033, TLI = 0.975, CFI = 0.981). Collectively, they demonstrate adequate model fit (Little, 2013). The results of the CFA are shown in Table B in the supplementary material. In addition, the CFA results were used to evaluate convergent and discriminant validity. The Bentler-Bonett (NFI) coefficient of 0.974 and statistically significant factor loadings ($p < 0.01$) indicate acceptable convergent validity. Discriminant validity was

evaluated using the heterotrait-monotrait (HTMT) method (Henseler *et al.*, 2015). This method supports discriminant validity since the results show HTMT values less than 0.70 for all constructs, with none of the confidence intervals including a value of one (Henseler *et al.*, 2015).

Experimental Analysis

Because our hypotheses suggest a moderated mediation model (see Figure 1), we followed established procedures (e.g., Hayes, 2013; Peinkofer *et al.*, 2016) using ordinary least square regression path analyses, which were conducted using the PROCESS function in RStudio. Based on Hayes (2013), 95% bias-corrected bootstrapped confidence intervals were calculated to test the indirect effects. None of the control variables were shown to have statistically significant effects when included in our models and thus were omitted from the presented analyses for increased clarity (Sturman *et al.*, 2022). To better differentiate the unique effects of logistics performance on impatience due to delivery speed and impatience due to timeliness, estimated models included both predictors and mediators.

The regression model results are presented in Tables I, II, and III. Hypothesis 1 predicted that impatience with delivery speed would mediate the relationship between delivery speed and consumer satisfaction. As shown in Table I, delivery speed was negatively related to impatience due to delivery speed ($\beta = -2.74$, $p < .01$). In turn, impatience due to delivery speed was negatively related to satisfaction ($\beta = -0.13$, $p = .01$). The indirect (mediation) effect was 0.36 with a 95% confidence interval of (.03, .70), supporting the proposed mediating effect. Thus, H1 was supported. H2 predicted a similar mediating effect for impatience with delivery timeliness. Delivery timeliness was negatively related to impatience due to timeliness ($\beta = -3.54$, $p < .01$), and impatience with timeliness was negatively related to satisfaction ($\beta = -0.26$, $p < .01$). The indirect effect was 0.92 with a 95% confidence interval of (.42, 1.42), supporting H2.

A moderated mediation model was used to test hypotheses 3 and 4. As highlighted in Table II, communication did not significantly moderate the relationship between impatience with speed and consumer satisfaction ($\beta = -0.07$, $p = .32$). The index of moderated mediation was estimated to be 0.03 with a 95% confidence interval of (-0.11, 0.17), suggesting a lack of moderated mediation (Hayes, 2015). Thus, H3 was not supported. Similarly, communication did not significantly moderate the relationship between impatience with timeliness and consumer satisfaction ($\beta = -0.03$, $p = .62$). Here, the index of moderated mediation was estimated to be 0.12 with a confidence interval that again included zero (-0.43, 0.68). H4 was not supported. The indirect effects with and without communication are presented in Table III.

----- Table I About Here -----

----- Table II About Here -----

----- Table III About Here -----

Study 2

Experimental Design

A second study was developed to explore how consumers would respond to early (relative to on-time or late) arrivals. Early deliveries provide an interesting extension to Study 1 because the development of impatience related to positive (early) and neutral (on-time) deliveries has yet to be explored. Study 2 further builds upon Study 1 by exploring if communication of the reasons behind a late delivery affects consumer satisfaction. Study 2 uses the same general setup as Study 1, where participants were provided with an online purchase scenario and asked to reflect on their experience through a subsequent survey. In Study 2, participants randomly assigned to

the “early” delivery timeliness manipulation experienced deliveries that arrived one day early, participants in the “on-time” manipulation experienced their deliveries on the originally promised date, and participants in the “late” manipulation had their deliveries arrive one day late. To allow us to more succinctly focus on the role of delivery timeliness, all scenarios utilized a delivery speed of three days and “high” levels of communication. Consistent with attribution theory, participants in the “late” scenario were also provided with one of two possible messages that indicated fault for the delivery delay (fault was either attributed to the shipper or the carrier). Our experimental design again attempted to indicate the passing of time by requiring participants to navigate through pages representing each day that they were waiting for their delivery, though the maximum number of pages was reduced relative to the first study due to the faster delivery speed employed. An overview of the scenarios and manipulations used in Study 2 is provided in the supplementary material.

Sample

A total of 300 responses were collected through MTurk, with 75 responses collected per scenario. Participants were excluded from the study for failing the timing or attention checks, resulting in a total of 265 responses in our analysis. The sample was 72.5% male; 93.2% identified their ethnicity as white. Much of the sample (75.8%) reported their age to be between 30 and 59 years. The annual income for 89.0% of participants was reported to be under \$100,000. The majority (83.3%) of participants indicated that they make online purchases at least once per month.

Measures

Our focal constructs were carried forward from study 1, but based on the helpful comments from anonymous reviewers, we adopted alternative measures to better triangulate our previous

findings and enhance the value of our internal replication (Rosenthal and Rosnow, 2008). Consumer satisfaction remained the primary outcome variable, and we adopted a single-item measure of overall satisfaction with the delivery experience that has been employed in other recent research (e.g., Ta *et al.*, 2023; Vakulenko *et al.*, 2022): “Overall, how satisfied were you with everything related to the delivery of this product” on a 7-point Likert scale ranging from (1) “extremely dissatisfied” to (7) “extremely satisfied.” The scale items for impatience due to delivery timeliness were adapted from Study 1 so that they could better capture a wider range of delivery timeliness outcomes: (1) “I feel frustrated that my order will not arrive on the guaranteed date”, (2) “I feel upset that the delivery date originally promised was incorrect”, (3) “I feel unhappy that the actual delivery date is different than what was originally promised”, (4) “I am concerned by the inaccuracy of the delivery timing”, and (5) “I feel anxious that I have not yet received my delivery.” Control variables in this study remained age, income, gender, and online purchase frequency. Descriptive statistics and correlations from Study 2 are summarized in Table D in the supplementary material.

Experimental Analysis

Before conducting our focal analysis, we performed an initial analysis to explore differences in satisfaction and impatience with timeliness across the different experimental conditions. We discovered that the means for impatience with timeliness ($M_{\text{Early}} = 5.00$, $M_{\text{OnTime}} = 4.73$; $t = -1.00$ ($df = 125$), $p = 0.317$) and satisfaction ($M_{\text{Early}} = 5.88$, $M_{\text{OnTime}} = 5.76$; $t = -1.02$ ($df = 125$), $p = 0.311$) were highly similar between the “early” and “on-time” conditions with no statistically significant differences. Similarly, regardless of whether the “lateness” was attributed to the retailer or the shipper, the mean level of satisfaction was nearly identical, with no significant difference ($M_{\text{Retailer}} = 5.50$, $M_{\text{Shipper}} = 5.42$; $t = 0.45$ ($df = 133$), $p = 0.65$). Based on these results,

we grouped the data into “on-time” (containing the original “early” and “on-time” conditions) and “late” (containing both retailer and shipper fault conditions) designations for subsequent analyses.

Paralleling the first study we used ordinary least square regression path analyses to evaluate whether impatience with delivery timeliness mediated the effect of delivery timeliness on consumer satisfaction in this experimental context. The regression results are shown in Table IV. Replicating the results of study 1 in this context, delivery timeliness was negatively related to impatience with timeliness ($\beta = -0.61$, $p < .01$) and impatience with timeliness was negatively related to satisfaction ($\beta = -0.16$, $p = .04$). Further, the indirect effect analysis shows that impatience with delivery timeliness significantly mediates the relationship between delivery timeliness performance and consumer satisfaction (effect size = 0.10, confidence interval = (.01, .23)). Consistent with Study 1, none of the control variables were found to have significant effects and were there omitted from the presented analysis to enhance interpretability of the focal results (Sturman, 2022).

----- Table IV About Here -----

Discussion

Making consumers happy is not an easy task – consumers expect not only high-quality products but also high-quality logistics performance (Daugherty *et al.*, 2019). LSCM research to date has mostly focused on consumer satisfaction as a major outcome variable of different logistics activities (Esper and Peinkofer, 2017). To better understand how firm supply chain practices can affect satisfaction outcomes, we answer a call to develop a deeper understanding of the

mechanisms that play a role in the development of consumer satisfaction (Esper *et al.*, 2020).

Utilizing attribution theory to develop middle-range theorising, we highlight the importance of considering consumer impatience related to time-based logistics performance.

The results of our data analysis reflect the important role of consumers' emotions prior to the receipt of a delivery in the development of satisfaction with delivery performance. We confirmed the existence and importance of consumer impatience in the context of e-commerce home deliveries. Specifically, our findings demonstrate that consumer impatience plays a mediating role in the relationship between time-based logistics performance and consumer satisfaction. This suggests that firms looking to make meaningful supply chain improvements should pay special attention to delivery speed and delivery timeliness performance (Del Valle, 2019; Loeb, 2021) with the goal of reducing consumer impatience and maximizing satisfaction outcomes. While the generic negative relationship between impatience and satisfaction seems intuitive, identifying and validating the sources of consumer impatience has important implications for supply chain managers since improvement efforts can be precisely focused on the most relevant logistics areas.

Our study goes beyond proving the relevance and importance of consumer impatience related to two time-based logistics capabilities – delivery speed and delivery timeliness. Our analysis indicates that consumers seem to value on-time deliveries more than delivery speed, based on the effects of their resultant impatience on satisfaction. This finding has crucial value in terms of guiding companies' efforts in improving their logistics performance: firms should focus on making sure deliveries arrive on time, even if that means indicating longer delivery lead times to their consumers. Faster delivery speed but missing promised delivery times ultimately result in lower levels of consumer satisfaction than slower deliveries that arrive on-time.

Study 2 expands the original conceptualization of timeliness by exploring the impact of early deliveries on the hypothesized relationships. While early deliveries in business-to-business (B2B) contexts can be negative, that doesn't seem to be the case in business-to-consumer (B2C). Collectively, our studies suggest that while there is value in ensuring orders arrive on time, there seems to be little value in providing early deliveries. Specifically, our results indicate that there is no significant difference in consumer impatience or satisfaction with early deliveries when compared to on-time deliveries. This finding should be valuable to firms working to effectively balance the tradeoffs between speed and timeliness when developing time-based delivery strategies.

Communication related to deliveries has a direct and positive relationship with consumer satisfaction. However, our findings do not support the moderating effects of delivery communication. There are several possible reasons for the insignificant moderating effects of delivery communication in this experiment. First, delivery communication could be unable to decrease the negative effects of impatience since consumers may view the focal firm as responsible for the entire fulfillment process. It is likely that consumers are attributing delivery issues to the focal firm since they were responsible for selecting the shipper, limiting the viability of a strategy of blaming an externality to reduce negative character attributions. Second, the lack of support for the moderating effect of delivery communication could be due to the timing and nature of consumer impatience. Once consumer impatience related to delivery performance is established, it might become ingrained in the minds of consumers. If that happens, it will be very difficult to change the negative emotions – even with a higher level of delivery communication. Third, the lack of support for communication as a moderator in this study could be related to the study design. There are many different styles of communication. However, within this research,

we only employed basic styles of email communication. One method of communication that could be explored in subsequent research is related to interactive retailer websites that can provide consumers with the ability to communicate with company representatives in real-time. Future studies could benefit from further exploration of communication as a moderator in the relationships between impatience related to delivery performance and satisfaction.

Conclusion

Our research provides a unique perspective on time-based competition and, more specifically, the importance of consumer impatience. By understanding that consumer impatience exists and plays an important role in the relationship between logistics performance and consumer satisfaction, we argue that companies will be better positioned to develop effective time-based delivery strategies, ultimately resulting in higher consumer satisfaction levels. We believe that our findings have meaningful implications for both research and practice.

Theoretical Implications

By theorising at the middle-range (Craighead *et al.*, 2024; Stank *et al.*, 2017), we are able to develop knowledge related to (1) the presence of consumer impatience in the relationship between time-based delivery performance and consumer satisfaction and (2) the impact of delivery speed and delivery timeliness on the emergence of consumer impatience in the e-commerce home delivery context. Additionally, we build upon previous research that has called for a better understanding of human-related factors in supply chain management processes (Bendoly *et al.* 2006), including the psychological conditions that consumers experience before they form their final opinions related to satisfaction (Esper *et al.*, 2020).

While previous research has conceptually discussed the presence of consumer impatience in a logistics service context (e.g., Daugherty *et al.*, 2019; Russo *et al.*, 2022), ours is the first to

empirically explore the presence of impatience related to e-commerce delivery performance. Demonstrating that impatience mediates the relationship between logistics service and consumer satisfaction provides an important contribution to the literature by extending the understanding of the nuance surrounding this well-established relationship. For example, we show that consumer impatience is directly tied to the occurrence of negative events in time-based delivery contexts. Our findings do not support the role of impatience in situations where deliveries can arrive “early” or “on time,” indicating that negative events are critical in the development of consumer impatience and its subsequent impact on satisfaction with delivery performance. It is important to note that while consumers typically do not view early deliveries as a negative event, early deliveries may be viewed differently in business-to-business transactions, which often involve carefully timed arrivals to and departures from heavily trafficked facilities.

Managerial Implications

From a managerial standpoint, our study suggests that companies will benefit from a better understanding of consumers’ emotions before their final satisfaction or dissatisfaction is formed. Proactively addressing emotions such as consumer impatience (i.e., negative emotions while waiting) will increase firms’ ability to achieve higher levels of consumer satisfaction. Specifically, the results of this study show that firms should focus their attention on ensuring that the lead times provided to their consumers are accurate. This might include investments in technology resources that support accurate delivery planning, allowing them to accurately set consumers’ lead time expectations at the point of sale and avoid late deliveries that result in impatience. It might also involve careful selection of external delivery carriers or management of internal fleets. While short lead times are important for competing in the marketplace, our work shows the value of on-time delivery regardless of the stated delivery speed. Further, our work

suggests that consumers seemingly do not care whether the late delivery is due to the shipper or the carrier, indicating that communication attempting to shift blame are unlikely to be successful in mitigating the negative consequences of late deliveries. Ensuring that delivery reliability is properly prioritized when selecting and evaluating carriers is critical for achieving consumer satisfaction in ways beyond what was previously understood. Adopting a focus on both delivery speed and timeliness will allow firms to directly impact the emergence of impatience related to e-commerce home deliveries.

Our findings suggest that firm efforts should be focused on (1) improving time-based logistics service areas that are most relevant to consumers; (2) setting realistic delivery expectations in the first place; and (3) understanding the limitations of communicating with consumers during the delivery process. Our study provides a new avenue to investigate means of achieving consumer satisfaction. Studying consumers' psychological states – consumer impatience in the current study – is certainly in line with the call for more consumer-centric approach to LSCM (Esper *et al.*, 2020). Positioning consumer impatience in the logistics context helps emphasize the importance of developing consumer-centric supply chains that firms can understand and support the needs of their consumers. Our study not only supports the notion that consumer impatience is related to logistics performance but also plays an important role in the development of consumer satisfaction. Addressing sources of consumer impatience will allow firms to positively influence consumer satisfaction. Our impatience research extends existing knowledge of how firms can develop logistics service strategies related to time-based competition in e-commerce settings.

Limitations and Future Research

Looking ahead, we believe our research takes an important step forward to incorporate consumer impatience into consumer-centric LSCM research. However, this study also has several limitations. We believe that these limitations can be used to pave the way for an abundance of future research opportunities. First, while we focus on two important areas – delivery speed and delivery timeliness – as the key sources of consumer impatience, we believe other delivery-related factors could also impact the emergence of consumer impatience. Therefore, monitoring and understanding additional LSCM-related delivery touchpoints that could act as sources of consumer impatience (e.g., delivery costs, delivery order accuracy, and the availability of a range of delivery options) could provide interesting topics for future research in this area (Vakulenko *et al.*, 2024). Second, while this study was conducted using a combination of attribution theory and middle range theorising, future LSCM research should continue pursuing theory-driven research opportunities that are capable of capturing consumer-centric phenomena. Specifically, researchers should work to employ a range of theorising approaches to develop context-specific insights that are critical to understanding consumer perceptions and behaviors (Baldi *et al.*, 2024). As one example, the rise of ecommerce retailers, such as Temu and Shein, offering ultra-low prices but notably longer delivery times relative to established competitors, raises intriguing questions about the role of additional purchase-related factors in the development of impatience. Various theoretical lenses could be used to guide future experiments designed to more fully understand observed consumer behavior. Third, the current study was limited to exploring the existence and impact of consumer impatience related to a specific set of delivery capabilities. However, a true consumer-centric supply chain involves many more companies and processes upstream than what were considered in this study. Future research would benefit from considering a wider variety of stakeholders involved in the e-commerce home-delivery process

including drivers, retailers, and service providers Masorgo *et al.* (2024). We encourage future research to explore other LSCM elements that may be related to consumer impatience (such as product or service development, manufacturing, operations, etc.) because a more holistic perspective of the supply chain is likely to generate more meaningful results that are relevant to practices. Fourth, our study focused on the business-to-consumer interface as the research context, but we expect that this phenomenon will have an important impact in the business-to-business context as well. Therefore, we propose that (1) future studies can be conducted in business-to-business settings to further investigate the impacts of consumer impatience on upstream supply chain processes and activities; and (2) the impatience concept can be studied in the business-to-business setting because the buyer-supplier relationships and interactions are likely to be affected by business customers' impatience too. Fifth, while we attempted to increase realism and immersion via the inclusion of pictures and the requirement to navigate through pages related to the number of days participants had to wait for their deliveries to arrive, this was but one way of signaling the passage of time. Future research should explore alternative means of operationalizing this phenomenon in the experimental design to further validate the generalizability of our findings.

In summary, we argue that consumer impatience as a consumer psychological state is a relevant but largely overlooked concept. Because it directly impacts consumer satisfaction, consumer impatience should be properly considered and managed. Effectively addressing consumer impatience is crucial to developing time-based logistics strategies, and thus we call for additional research surrounding this important topic.

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Table I. Study 1 mediation model results for consumer impatience predicting consumer satisfaction

Predictor	Outcome					
	S_Impatience		T_Impatience		Satisfaction	
	β (SE)	p	β (SE)	p	β (SE)	p
Intercept	6.13 (.14)	.00	6.20 (.12)	.00		
Delivery speed	<u>-2.74 (.16)</u>	.00	-0.87 (.15)	.00		
Delivery timeliness	-1.38 (.16)	.00	<u>-3.54 (.15)</u>	.00		
Intercept					4.21 (.37)	.00
Delivery speed					0.83 (.19)	.00
Delivery timeliness					1.31 (.23)	.00
S_Impatience					<u>-0.13 (.05)</u>	.01
T_Impatience					<u>-0.26 (.06)</u>	.00
R^2	.50		.63		.54	
F-Statistic	185.0 (p < .01)		320.0 (p < .01)		110.8 (p < .01)	

Source(s): Table created by authors

Note(s): *S_Impatience = Impatience due to delivery speed, T_Impatience = Impatience due to delivery timeliness. Results relevant for testing H1 and H2 are underlined. Standard errors are presented in parentheses following the parameter estimates. p-values displayed as .00 are significant at the 0.01 level.*

Table II. Study 1 moderated mediation model results for consumer impatience predicting consumer satisfaction as moderated by communication

Predictor	Outcome					
	S_Impatience		T_Impatience		Satisfaction	
	β (SE)	<i>p</i>	β (SE)	<i>p</i>	β (SE)	<i>p</i>
Intercept	6.13 (.14)	.00	6.20 (.12)	.00		
Delivery speed	-2.74 (.16)	.00	-0.87 (.15)	.00		
Delivery timeliness	-1.38 (.16)	.00	-3.54 (.15)	.00		
Intercept					2.80 (.39)	.00
Delivery speed					0.93 (.17)	.00
Delivery timeliness					1.66 (.21)	.00
S_Impatience					-0.10 (.06)	.08
T_Impatience					-0.14 (.06)	.02
Communication					1.61 (.30)	.00
S_Impatience* Communication					<u>-0.07 (.07)</u>	.32
T_Impatience* Communication					<u>-0.03 (.07)</u>	.62
R^2	.50		.63		.62	
F-Statistic	185.0 ($p < .01$)		320.0 ($p < .01$)		87.6 ($p < .01$)	

Source(s): Table created by authors

Note(s): *S_Impatience* = Impatience due to delivery speed, *T_Impatience* = Impatience due to delivery timeliness. Results relevant for testing H3 and H4 are underlined. Standard errors are presented in parentheses following the parameter estimates. *p*-values displayed as .00 are significant at the 0.01 level.

Table III. Study 1 conditional indirect effects of time-based delivery performance on consumer satisfaction through consumer impatience at different levels of communication

Delivery Speed	Comm.	$\theta_{(DS \rightarrow IMP_DS)} = a_1$	$\theta_{(DS \rightarrow IMP_DS)} = b_1$	Indirect		
				Effect Size	BootLL	BootUL
	0	-2.74	-0.10	0.27	-0.06	0.62
	1	-2.74	-0.17	0.47*	0.04	0.91
Delivery Timeliness	Comm.	$\theta_{(DT \rightarrow IMP_DT)} = a_1$	$\theta_{(DT \rightarrow IMP_DT)} = b_1$	Indirect		
				Effect Size	BootLL	BootUL
	0	-3.54	-0.14	0.51*	0.03	0.98
	1	-3.54	-0.18	0.63*	0.05	1.21

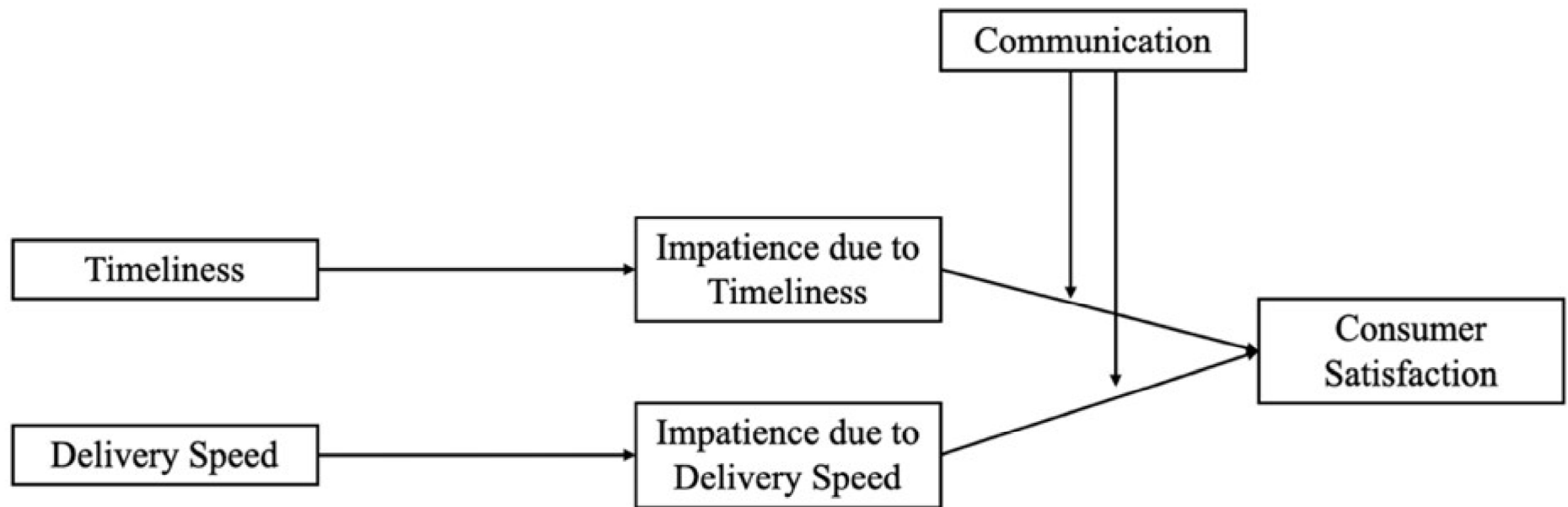
Source(s): Table created by authors

Note(s): * Significant since 0 is not included in the 95% Bootstrap confidence interval; DS = Delivery speed; S_Impatience = Impatience due to delivery speed; DT = Delivery timeliness; T_Impatience = Impatience due to delivery timeliness.

Table IV. Study 2 mediation model results for consumer impatience predicting consumer satisfaction

Predictor	Outcome			
	T_Impatience		Satisfaction	
	β (SE)	p	β (SE)	p
Intercept	5.45 (.10)	.00		
Delivery timeliness	<u>-0.61 (.15)</u>	.00		
Intercept			5.68 (.46)	.00
Delivery timeliness			0.70 (.20)	.00
T_Impatience			<u>-0.16 (.08)</u>	.04
R^2	.06		.07	
F-Statistic	16.75 ($p < .01$)		10.32 ($p < .01$)	
Source(s): Table created by authors				

Note(s): *T_Impatience = Impatience due to delivery timeliness. Values relevant to assessing the mediating effects of impatience with timeliness are underlined. p -values displayed as .00 are significant at the 0.01 level.*



Source(s): Figure created by the authors

SUPPLEMENTARY MATERIAL: TABLES

Table A. Recent studies exploring the relationship between time-based delivery performance and consumer satisfaction

Study	Key findings	Relative contribution of the present research
Cui <i>et al.</i> (2024)	- A faster delivery speed promise increases sales and profits, but it also increases product returns and reduces customer retention. - Retailers should optimize and customize their delivery promises to maximize customer lifetime value.	- Explores the impact of both delivery speed and delivery timeliness promises on consumers, contributing to this stream of research by providing an explanation as to why these time-based consumer preferences exist.
Gawor and Hoberg (2018)	- On average, lead time is a critical factor for consumers when selecting an omni-channel retailer. - Omni-channel retailers should focus on efficient B2C e-fulfillment strategies.	- Explores how firms can best develop efficient B2C e-fulfillment strategies by looking at the impact of different time-based strategies on consumer psychological responses that impact their satisfaction with a delivery experience.
Nguyen <i>et al.</i> (2019)	- Last-mile consumers tend to value delivery fees more than delivery speed. - Time- and convenience-oriented consumers will prioritize delivery speed and convenience when choosing a delivery option with multiple attributes.	- Explains why some consumer delivery preferences in online retailing might exist. - Includes delivery timeliness as a factor that impacts consumer preferences in online delivery settings.
Olsson <i>et al.</i> (2022)	- Delivery lead time is a critical factor for consumer adaptation of emerging last-mile delivery options in e-retail.	- Extends research into the dynamic and holistic nature of consumers' delivery experiences. - Takes a quantitative approach to studying a negative cognitive consumer response (impatience) to last-mile delivery services.
Song <i>et al.</i> (2024)	- Customers award higher ratings to orders that are delivered earlier in the day, regardless of the presence of a service guarantee or how many days it takes for the customer to receive an order.	- Presents evidence of a psychological factor that might explain consumer preferences.
Source(s): Table created by authors		

Table B. Confirmatory factor analysis

	Estimated Factor Loadings	S.E.	z-value	$p(> z)$	Cronbach's Alpha
<i>Impatience with Delivery Speed</i>					0.97
DS1: <i>I feel upset by how slow the guaranteed delivery speed was.</i>	2.03	0.09	22.77	0.00	
DS2: <i>I am displeased that there was no option for faster delivery.</i>	2.15	0.09	23.67	0.00	
DS3: <i>I feel unhappy that the initially promised delivery date for the order was far away.</i>	2.25	0.09	25.88	0.00	
DS4: <i>I am frustrated with the number of days before my order was supposed to arrive.</i>	2.31	0.09	25.94	0.00	
<i>Impatience with Delivery Timeliness</i>					0.97
T1: <i>I feel upset about having to wait for my late delivery to arrive.</i>	2.16	0.09	24.40	0.00	
T2: <i>I feel frustrated that my order did not arrive by the guaranteed date.</i>	2.38	0.09	25.76	0.00	
T3: <i>I feel unhappy that the delivery date is further away than originally promised.</i>	2.37	0.09	25.97	0.00	
T4: <i>I am concerned by the lateness of the delivery.</i>	2.27	0.09	24.98	0.00	
<i>Consumer Satisfaction</i>					0.98
CS1: <i>I am very satisfied with the service I received.</i>	1.90	0.08	24.14	0.00	
CS2: <i>My shopping experience was superior.</i>	2.03	0.08	25.56	0.00	
CS3: <i>I received very close to “perfect” service.</i>	2.08	0.08	25.13	0.00	
CS4: <i>This online retailer was able to offer superior service.</i>	2.06	0.08	25.14	0.00	
Comparative Fit Index (CFI)			0.981		
Tucker-Lewis Index (TLI)			0.975		
Root Mean Square Error of Approximation			0.086		
Chi-square			0.000		
Number of observations (n)			376		
Degrees of freedom			51		
Source(s): Table created by authors					

Table C. Study 1 scale statistics and correlations

Construct	Mean	SD	(1)	(2)	(3)	(4)	(5)	(6)
(1) Consumer Satisfaction	3.669	2.042	(.98)					
(2) Delivery Communication	0.495	0.501	.317*	-				
(3) Delivery Speed	0.500	0.501	.365*	-.016	-			
(4) Delivery Timeliness	0.481	0.500	.607*	-.017	.043	-		
(5) S_Impatience	4.094	2.219	-.554*	-.050	-.638*	-.337*	(.97)	
(6) T_Impatience	4.064	2.320	-.676*	-.121	-.223*	-.776*	.569*	(.97)
Source(s): Table created by authors								

Note(s): *Correlations are significant at the 0.01 level; n=374; S_Impatience = Impatience due to delivery speed, T_Impatience = Impatience due to delivery timeliness. Delivery speed coded such that 0 = 7 day expected delivery and 1 = 1 day expected delivery. Delivery timeliness coded such that 0 = late delivery and 1 = on-time delivery.

Table D. Study 2 scale statistics and correlations

Construct	Mean	SD	(1)	(2)	(3)
(1) Consumer Satisfaction	5.19	1.64	(.78)		
(2) Delivery Timeliness	0.48	0.5	.24*	-	
(3) T_Impatience	5.16	1.25	-.17*	-.25*	(.89)
Source(s): Table created by authors					

Note(s): T_Impatience = Impatience due to delivery timeliness. Delivery timeliness coded such that 0 = late delivery and 1 = on-time delivery. *Correlations are significant at the 0.01 level; n=265

SUPPLEMENTARY MATERIALS: STUDY 1 AND STUDY 2 RESEARCH DESIGN

1. All participants in Study 1 and Study 2 were provided with the following background information

We're asking you to take on the role of a consumer who has just completed an online purchase of a coffee maker.

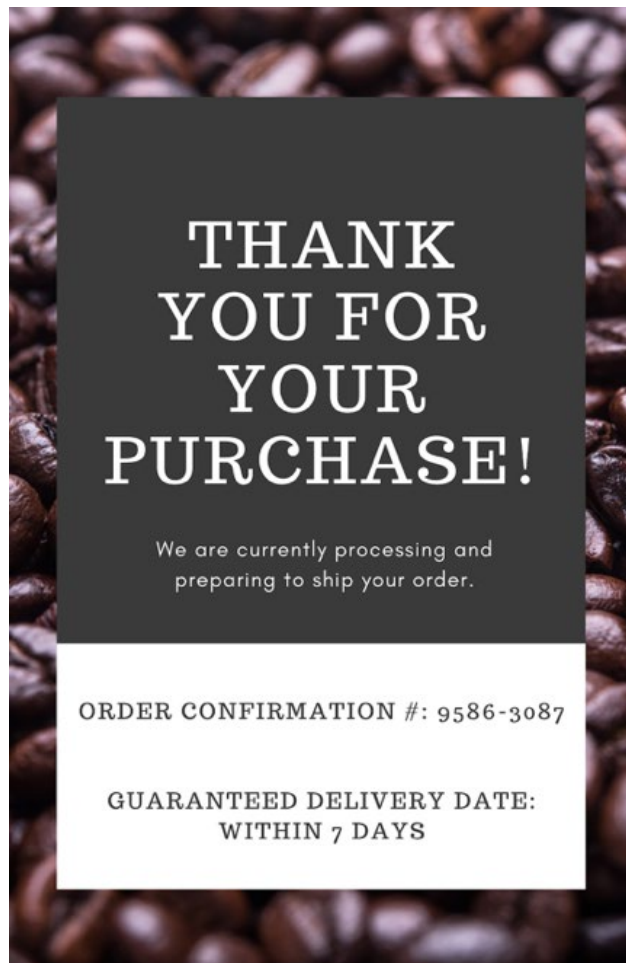
You are purchasing this coffee maker for yourself. Your old coffee maker broke earlier today. It is important that you get a replacement for your broken coffee maker as soon as possible. Because a fresh cup of coffee is an important start to your day, you select an express shipping option when placing your order.

The new coffee maker is identical to your old one. After placing your order at an online retailer, you pay a total of \$69.99. This cost includes the coffee maker, sales tax, and shipping and handling charges. This expense was not part of your monthly budget and is a little more than expected, but the cost is worth the benefit of having a fresh cup of coffee in the morning. The same coffee maker is available at the same price from other online retailers. The retailer that you have selected has positive online reviews.

Deliveries are made to your front door. You have just placed your order. You have not been provided with a delivery date. However, the website has informed you that the delivery details will arrive via email within a few minutes.

2. Study 1 Experiment Design

Low Delivery Speed Manipulation



Source(s): Figure created by authors

Figure A. Study 1 low delivery speed manipulation

“Your order has been guaranteed to be delivered within the next 7 days. Your new coffee maker can’t arrive soon enough.”

High Delivery Speed Manipulation



Source(s): Figure created by authors

Figure B. Study 1 high delivery speed manipulation

“Your order has been guaranteed to be delivered within 1 day. Your new coffee maker can’t arrive soon enough.”

“On-Time” Delivery Timeliness Manipulation

“By the end of the day, **your delivery has arrived.**”

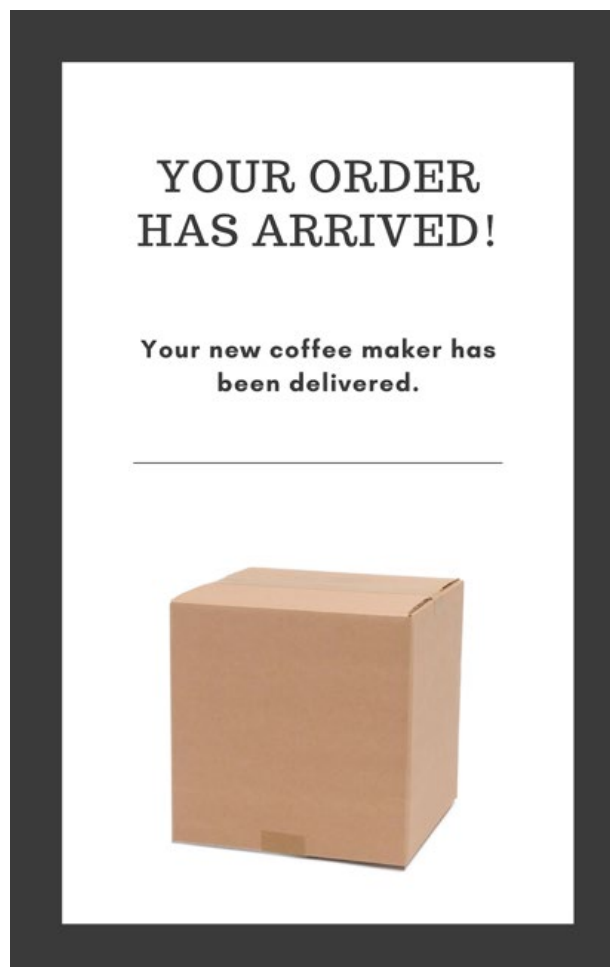
“Late” Delivery Timeliness Manipulation

“By the end of the day, **your delivery has not arrived.**”

The participant was then required to wait two additional days prior to the arrival of their delivery

Low Delivery Communication Manipulation

- Participants received an email confirming that their order had shipped and providing them with a guaranteed delivery date (shown as part of the delivery speed manipulation).
- While waiting for their delivery to arrive, participants were informed that “You do not have the ability to track your delivery”
- Near the end of the experiment, participants in this manipulation were given the following reminder: “**You have not been provided with any shipment updates or tracking information from the online retailer.**”
- Participants received an email at the end of the experiment confirming that their order had been delivered:

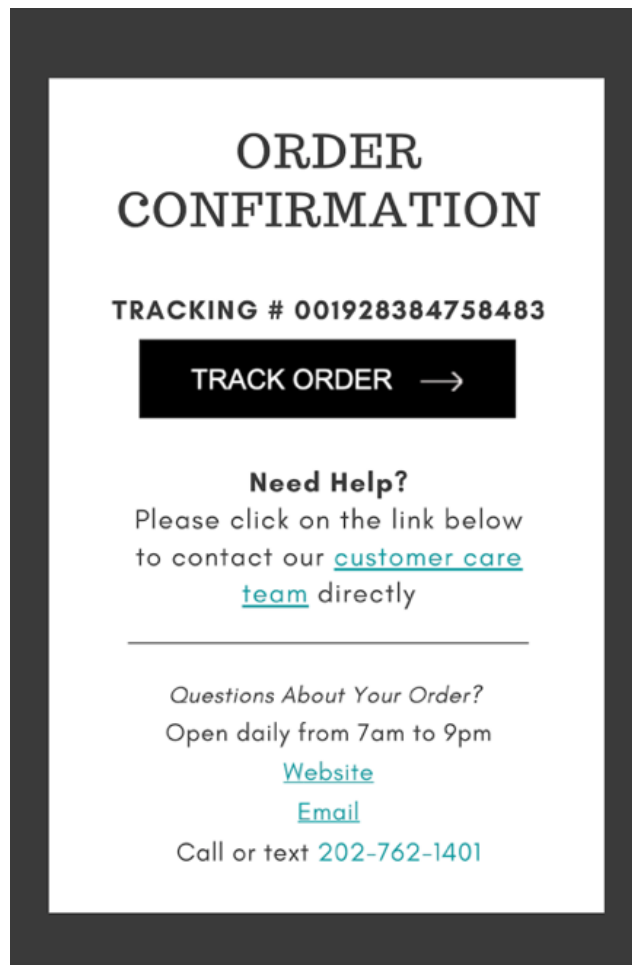


Source(s): Figure created by authors

Figure C. Study 1 delivery confirmation email

High Delivery Communication Manipulation

- Participants received an email confirming that their order had shipped and providing them with a guaranteed delivery date (shown as part of the delivery speed manipulation).
- In addition to the original order confirmation email, participants were also provided with an order confirmation message that provided a tracking number and the ability to contact a customer support team.



Source(s): Figure created by authors

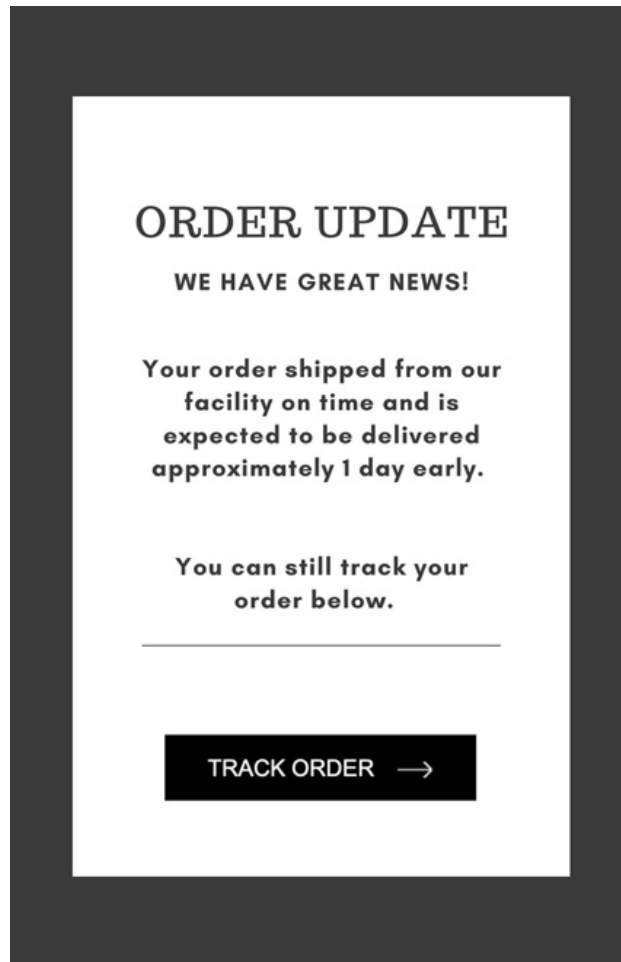
Figure D. Study 1 order confirmation email

- While waiting for their delivery to arrive, participants were informed that “You have the ability to track your delivery via the link provided in the order confirmation email.”
- Near the end of the experiment, participants in this manipulation were given the following reminder: “**You have been provided with several shipment updates and the ability to track the delivery of your order through the online retailer.**”
- Participants received an email at the end of the experiment confirming that their order had been delivered (This is the same email that was received at the end of the low delivery communication manipulation).

3. Study 2 Experiment Design

“Early” Delivery Timeliness Manipulation

“On the morning of Day 2, you receive an email with the following update:”

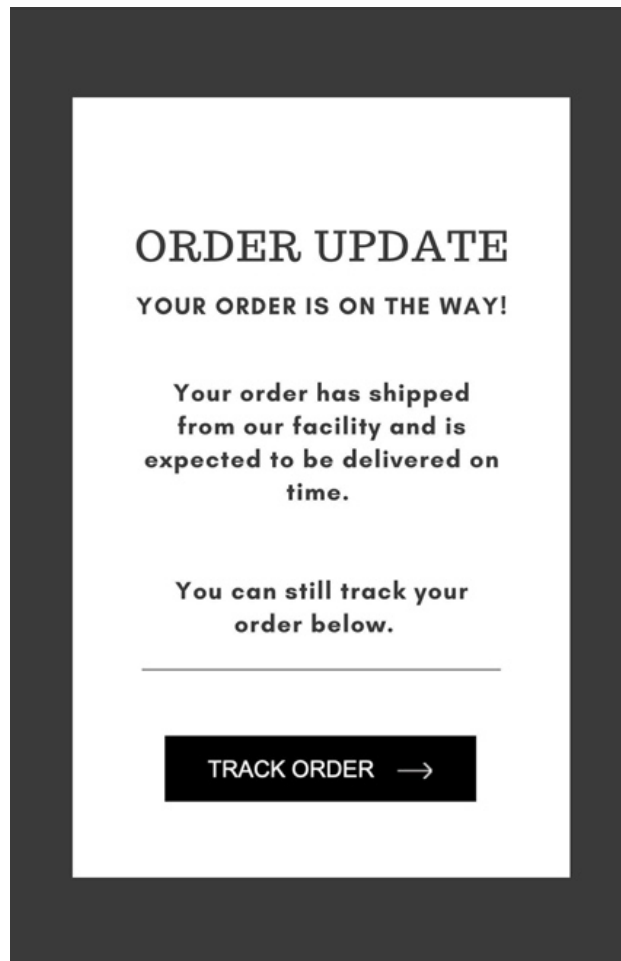


Source(s): Figure created by authors

Figure E. Study 2 “early” delivery timeliness manipulation

“On-Time” Delivery Timeliness Manipulation

“On the morning of Day 2, you receive an email with the following update:”

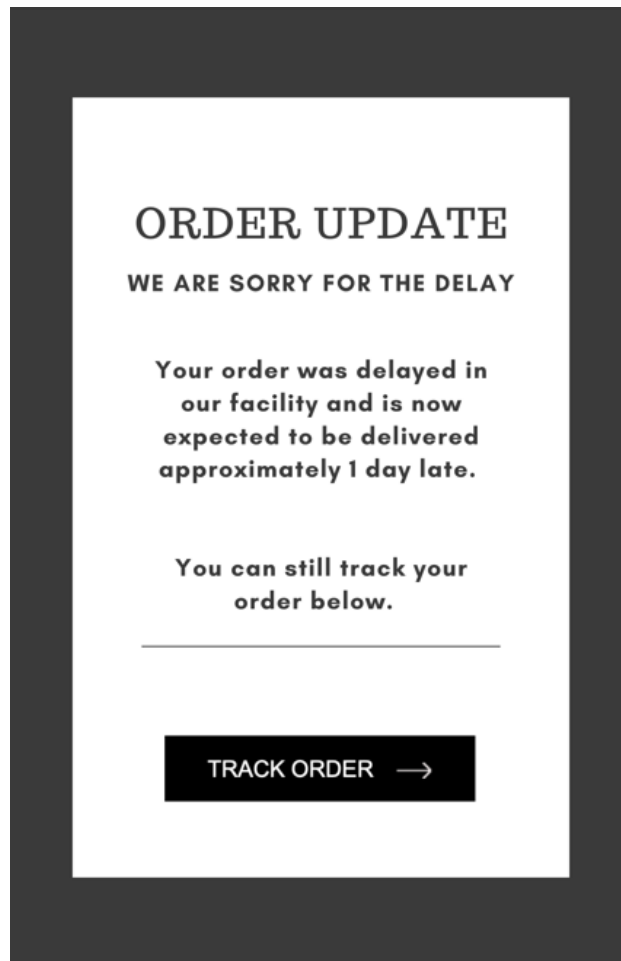


Source(s): Figure created by authors

Figure F. Study 2 “on-time” delivery timeliness manipulation

“Late – Due to the retailer” Delivery Timeliness Manipulation

“On the morning of Day 2, you receive an email with the following update:”



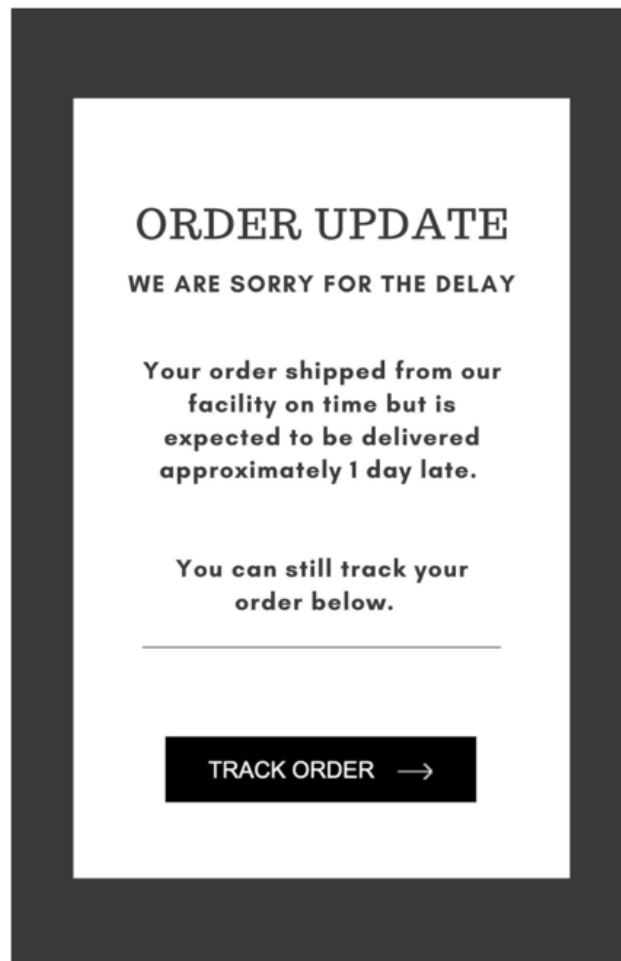
Source(s): Figure created by authors

Figure G. Study 2 “late – due to the retailer” delivery timeliness manipulation

The participant was then required to wait one additional day prior to the arrival of their delivery.

“Late – Due to the shipper” Delivery Timeliness Manipulation

“On the morning of Day 2, you receive an email with the following update:”



Source(s): Figure created by authors

Figure H. Study 2 “late – due to the shipper” delivery timeliness manipulation

The participant was then required to wait one additional day prior to the arrival of their delivery.

Delivery Speed

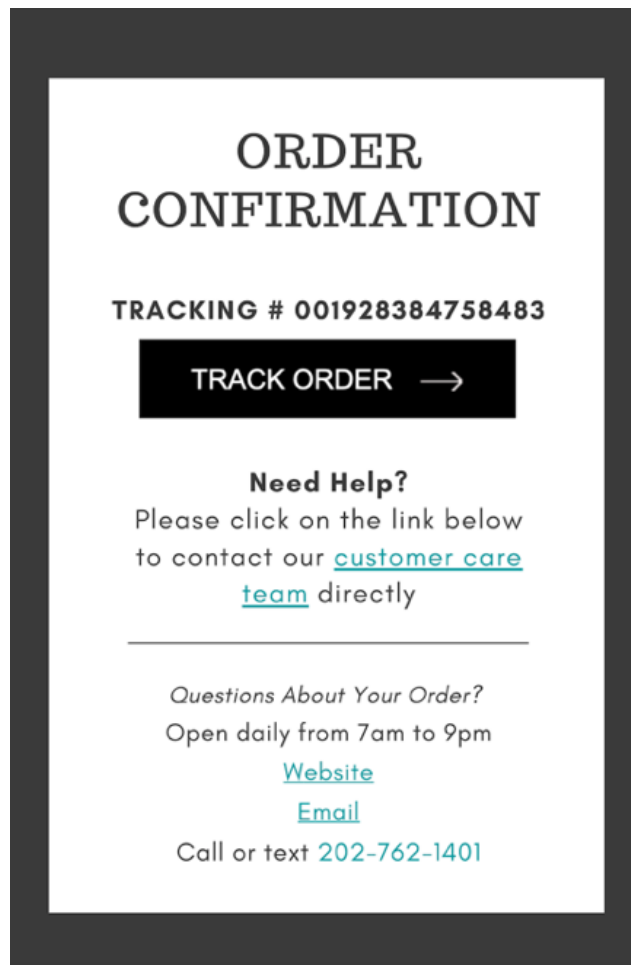
Source(s): Figure created by authors

Figure I. Study 2 delivery speed manipulation

“Your order has been guaranteed to be delivered within the next 3 days. Your new coffee maker can’t arrive soon enough.”

Delivery Communication

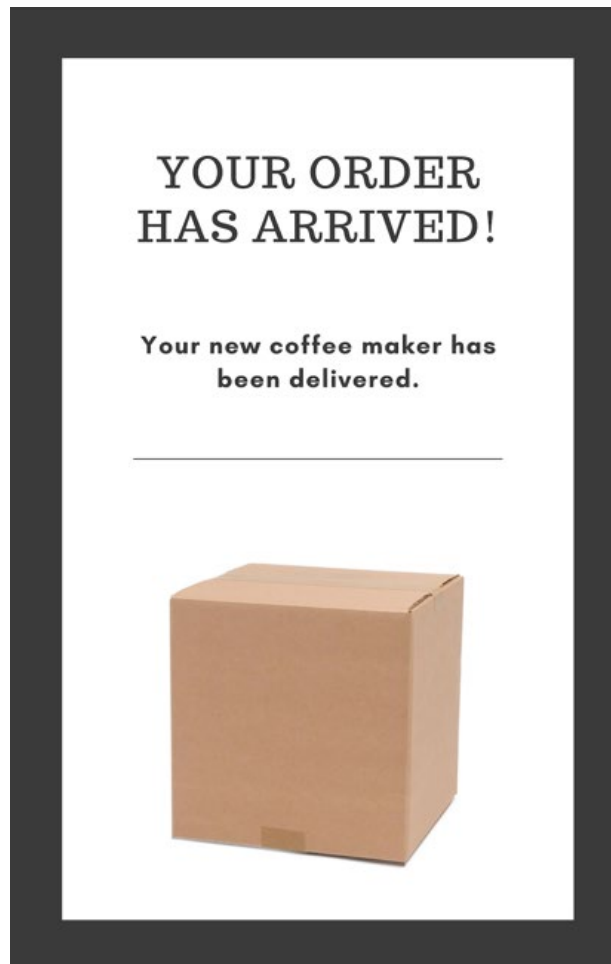
- Participants received an email confirming that their order had shipped and providing them with a guaranteed delivery date (shown as part of the delivery speed manipulation).
- In addition to the original order confirmation email, participants were also provided with an order confirmation message that provided a tracking number and the ability to contact a customer support team.



Source(s): Figure created by authors

Figure J. Study 2 order confirmation email

- While waiting for their delivery to arrive, participants were informed that “You have the ability to track your delivery via the link provided in the order confirmation email.”
- Near the end of the experiment, participants in this manipulation were given the following reminder: “**You have been provided with several shipment updates and the ability to track the delivery of your order through the online retailer.**”
- Participants received an email at the end of the experiment confirming that their order had been delivered.



Source(s): Figure created by authors

Figure K. Study 2 delivery confirmation email